Silicon & Handset — Supplier Ecosystem Simulation, Computational Fidelity & Cross-Seed World Divergence Blueprint

Target Audience: Autonomous Implementation Models (`GPT Sol 6.1`, `Gemini 4 Argon`, `Codex`, `Agy`)
Target Branch / Codebase: `spinoff-phone` (`SPINOFF_SOURCE/`)
Core Objective: Define **(1) the exact computational depth ("how much math to simulate")** for every component supplier category (`CPU/SoC`, `DRAM`, `NAND Storage`, `Camera/Optics`, `Display`, `Battery/BMS`, `Enclosure/Thermal`, `PCB/EMS`), and **(2) the procedural NPC + resource genesis architecture** that makes every new world seed evolve a genuinely different industrial history—driven 100% by randomized NPC capabilities (`HumanCore`), firm capital/equipment, and finite resources.
Non-Negotiable Invariants (`AGENTS.md`):

- Never modify `AGENTS.md`, `SPINOFF_SOURCE/js/human-core.js` (SHA-256 `fa83eaa0ef3407f26f21675161f632a4365242f67fea7b0c192e59e75b9d5c32`), or `SPINOFF_SOURCE/CORE_LOCK.json`.
- Zero scripted "world modifiers" or player-relative rubber-banding: a world where DRAM leaps ahead while CPUs stagnate must emerge **strictly** from the seeded NPCs, tools, capital, and market feedback of those firms.
- Strict integer satang conservation (`audit(w)`), physical lot conservation, partition-invariant hourly ticks, read-only UI purity, and natural Thai observer explanations.

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1. Computational Fidelity Contract: "How Deep Should the Math Go?"

1.1 The Golden Rule of Meso-Scale Simulation

- **Too Shallow (Current `v0.2.0`):** A single scalar `capability: 0..3` per supplier. Result: zero specialization, identical components, zero replayability.
- **Too Deep (Over-Engineered Trap):** Simulating Verilog gate timing, chemical lithography diffusion PDEs, or 40 sub-materials per capacitor. Result: slow ticks, opaque noise that players cannot observe, and brittle AI heuristics.
- **Optimal Meso-Scale Target (Budget: `< 0.4 ms` per simulated hour for 20+ firms):** Every component category is represented by **4–6 orthogonal physical parameters**, **1**

strict interface/compatibility contract, and **2 manufacturing economics variables** (`lotYield` and `unitDirectCostSatang`), evaluated via **closed-form algebraic physics equations**. Every parameter MUST directly affect at least one downstream handset benchmark, compatibility gate, thermal/power budget, or failure rate—**never add cosmetic numbers that nothing reads**.

1.2 Exact Per-Supplier Calculation Matrix

Supplier Domain	Explicit State & SKU Parameters to Compute	Closed-Form Physics & Yield Equations	What to Abstract Away (Do NOT Simulate)
1. CPU / SoC (`chip`)	`nodeNm`, `cores`, `ipcIndex`, `clockGhz`, `gpuGflops`, `ispMaxMp`, `npuTops`, `memBus`, `storageBus`, `dieAreaMm2`, `idleW`, `loadW`, `peakW`	• `dieAreaMm2` $\propto (\text{transistorBudget}) \cdot (\text{nodeNm}/130)^{1.35} / \text{layoutSkill}$ • `loadW` $\propto \text{cores} \cdot \text{ipc}^{1.1} \cdot \text{clockGhz}^{2.2} \cdot (\text{nodeNm}/130)^{0.75}$ • Murphy wafer yield `E.wafer(dieAreaMm2, D0)` $\rightarrow$ `A-Bin` / `B-Bin` / `scrap`	Instruction pipelines, cache coherency protocols, individual EUV mask layers
2. Volatile Memory (`ram`)	`memBus` (`SDRAM`..`LPDDR5`), `capacityMB`, `bandwidthGBps`, `latencyNs`, `idleRefreshMw`, `activeMw`	• `bandwidthGBps` $= \text{busClock} \cdot \text{busWidthBytes} \cdot \text{efficiency}$ • `idleRefreshMw` $\propto \text{capacityMB} \cdot$	Row-hammer bit flips, sub-bank refresh timing state machines

Supplier Domain	Explicit State & SKU Parameters to Compute	Closed-Form Physics & Yield Equations	What to Abstract Away (Do NOT Simulate)
		\text{leakageFactor}(\text{circuitSkill})$ • Yield $\propto \exp(-\text{densityFactor} \cdot D_{0,\text{dram}})$	
3. Flash Storage (`storage`)	`storageBus` (`SPI`/`eMMC`/`UFS`), `cellBits` (`1=SLC..4=QLC`), `capacityMB`, `seqReadMBps`, `seqWriteMBps`, `randIops`, `enduranceCycles`	• Higher `cellBits` lowers cost/MB by $1/\text{cellBits}^{0.75}$ but cuts `seqWriteMBps` & `enduranceCycles` exponentially unless compensated by `embedded_firmware` controller ECC skill	Flash translation layer block tables, page-level garbage collection ticks
4. Camera & Optics (`camera`)	`sensorAreaMm2`, `megapixels`, `pixelPitchUm`, `apertureF`, `lensElements`, `hasOis`, `afType`, `zHeightMm`	• `pixelPitchUm` $= \sqrt{\text{sensorAreaMm2} / \text{megapixels}} \cdot 1000$ • `photonIndex` $\propto (\text{pixelPitchUm}^2 / \text{apertureF}^2) \cdot (1 + 0.25 \cdot \text{hasOis})$ • Cleanroom optical alignment yield $\propto f(\text{opticsSkill}, \text{lensElements})$	Ray-tracing lens distortion curves, Bayer demosaicing pixel arrays
5. Display Panel (`display`)	`panelTech` (`TFT`/`IPS`/`OLED`/`LTPO`	• `activeWatts` $= \text{areaFactor}	Sub-pixel pentile geometry, color

Supplier Domain	Explicit State & SKU Parameters to Compute	Closed-Form Physics & Yield Equations	What to Abstract Away (Do NOT Simulate)
	), `interface`, `width`, `height`, `peakNits`, `refreshHz`, `wattsAt200Nits`, `contrastScore`	\cdot \text{nitsFactor} \cdot \text{refreshFactor} / \text{panelEff}$ • Mother-glass panel cut yield $\propto \exp(-\text{panelArea} \cdot D_{0,\text{panel}})$	gamut CIE coordinates
6. Battery & BMS (`battery`)	`chemistry` (`LiIon`/`LiPo`/`SiCarbon`), `capacityWh`, `volumeCm3`, `maxDischargeW`, `maxChargeW`, `cycleLife80`, `bmsSafetyScore`	• `capacityWh` $= \text{volumeCm3} \cdot \text{energyDensity}(\text{chemistry}, \text{powerSkill})$ • Fast-charge thermal wear $\propto (\text{maxChargeW}/\text{capacityWh})^{1.4} / \text{bmsSafetyScore}$	Electrochemical ion diffusion PDEs, voltage discharge curves per minute
7. Enclosure & Thermal (`enclosure`)	`material`, `coolingTech`, `internalVolumeCm3`, `maxModuleZMm`, `rthCW`, `thermalMassJC`, `thicknessMm`, `weightG`, `dropScore`, `rfAttenuationDb`	• $T_{\text{skin}} = T_{\text{amb}} + P_{\text{peak}} \cdot \text{rthCW}$ • Metal raises `dropScore` & lowers `rthCW`, but increases `rfAttenuationDb` unless mitigated by precision CNC antenna slots (`machiningSkill`)	3D CAD mesh geometry, finite-element CFD airflow simulation

Supplier Domain	Explicit State & SKU Parameters to Compute	Closed-Form Physics & Yield Equations	What to Abstract Away (Do NOT Simulate)
8. EMS / OSAT / PCB (`ems`)	`maxBoardLayers`, `signalIntegrity`, `smtYield`, `setupFeeSatang`, `perUnitFeeSatang`, `lineCapacityUnitsPerDay`	• Batch unit assembly cost $= \text{perUnitFee} + \text{setupFee} / \text{batchQty}$ (Economies of Scale!) • Assembly defect scrap $= 1 - \text{smtYield}(\text{manufacturingSkill}, \text{complexity})$	Individual pick-and-place nozzle trajectories, solder paste reflow curves

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2. Procedural World & NPC Genesis: Why Every Seed Must Evolve Differently

2.1 Why Current Seeds Converge (The Central-Limit Trap to Avoid)

In `v0.2.0`, every supplier spawns with `3..5` employees whose skills are drawn from narrow uniform ranges (`34..76`), `40M..80M` satang cash, and identical access to infinite `1,200`-satang upstream material. Averaging 4 uniform draws per firm forces every sector toward the same mean (~55), causing all 6 component sectors to advance at nearly identical speeds in every playthrough.

2.2 Heavy-Tailed Procedural Genesis (`createWorldEcosystem(seed)`)

Use isolated deterministic PRNG streams (`hash(seed + ':genesis:' + domain)`) to generate **authentic structural asymmetry** at `minute = 0` without any artificial "buffs":

Mechanism A — Procedural NPC Roster with Career Archetypes & Rare Luminaries

Generate every person via immutable `HumanCore.createProfile(seed, personId)` first, then derive their **starting career stage & learned `subjects`** from their actual `HumanCore` potentials (`H.skillCap(profile, SUBJECT_SPECS[s])`), age/experience roll, and prior background:

1. **Junior / Apprentice (25% of pool):** Young talent; low current `subjects` (`20..42`), lower initial wage demand, potentially high `HumanCore potential` + `learningSpeed` (diamond in the rough!).
2. **Journeyman Engineer (50% of pool):** Solid working specialists (`subjects 40..64` in 1–2 core fields).
3. **Senior Specialist (20% of pool):** Experienced veterans (`subjects 62..78`, higher wage expectations).
4. **Generational Domain Luminary (~5% tail probability per roll, gated by actual `HumanCore` aptitude fit `>= 78`):** A rare, authentic outlier whose `HumanCore analytical`/`precision`/`visualAcuity` potential is `82..96` AND who starts with deep prior domain mastery (`78..92`) in *one specific discipline* (e.g., `dram_circuit`, `semiconductor_lithography`, `optics_design`, or `electrochemistry`), demanding a high salary and holding strong `HumanCore` personality convictions (`openness`, `ambition`, `craftsmanship`, or `autonomy`).

- **Emergent Result Across Seeds:**
    - In **Seed A**, the random NPC draft happens to place a Generational Luminary in `DRAM` (`bandwidth`/`LPDDR` pioneer) and two Senior `Optics` specialists in `Camera`, while `CPU` firms only rolled Journeymen and a cautious, risk-averse CEO!
    - In **Seed B**, a small `Chip` startup rolls a brilliant high-`riskTaking` VLSI architect + an `OSAT` firm rolls a precision yield master, while `Battery` and `Display` firms start with inexperienced teams!

Mechanism B — Asymmetric Firm Founding Archetypes (Capital × Equipment × Team Size)

Instead of giving every supplier `~50M` satang and `3..5` staff, assign each seeded supplier one of four realistic industrial founding structures rolled from the world seed:

1. **Lean Specialist Boutique (2–3 staff, low cash, specialized precision rig):** Low burn rate; excels at high-margin niche design (e.g., fabless IP or boutique optics), vulnerable to cash crunches if a prototype fails.
2. **Well-Capitalized Industrial Spin-Out (5–7 staff, high cash reserve, mature line equipment):** High daily wage burn; can fund expensive NRE/tooling early, but goes bankrupt fast if bureaucratic/cautious leadership (`low openness`, `high prudence`) sits on obsolete SKUs.
3. **High-Volume Commodity Producer (4–6 staff, high `manufacturing` skill, high raw stock, modest R&D skill):** Dominates cheap, high-yield mature lots (`SDRAM`, `IPS`, `Polymer Chassis`) through scale and low prices.
4. **Unfunded Moonshot Lab (3–4 high-potential engineers, tight cash runway, ambitious CEO):** Races to ship a breakthrough SKU before its 60-day wage runway runs out—creating dramatic early-game booms or talent-scattering bankruptcies!

Mechanism C — Seeded Upstream Resource & Regional Infrastructure Endowments

Vary the world's starting upstream material availability and base extraction costs (`w.upstreamMarkets` replacing a single undifferentiated `raw` bucket with **4–5 bounded industrial input families**: `silicon_wafer_feed`, `optical_glass_chemical`, `display_substrate`, `battery_cell_ precursor`, `structural_alloy_polymer`):

- Each seed rolls distinct base extraction capacities and unit costs (`0.75x .. 1.45x`) across those 5 upstream feeds (representing regional supply strengths/bottlenecks).
- A world with abundant, cheap `silicon_wafer_feed` and scarce, expensive `battery_cell_precursor` naturally accelerates semiconductor experimentation while forcing the entire phone industry to obsess over power efficiency!

Mechanism D — Seeded Unemployed & Emerging Talent Stream

Populate an initial pool of `10–16` diverse unemployed specialists (`outside` pool) plus periodic **University Graduate / Industry Entrant** arrivals (bounded population cap, e.g., `max 120` people total, funded via conservation-exact household transfers on hiring) so growing winners can expand headcount (`up to 10–12` staff) if they earn the cash to pay competitive wages.

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3. Autonomous Supplier Brain: How Each Vendor Chooses R&D, Production & Pricing

Every supplier firm runs three autonomous decision loops without any player intervention:

3.1 R&D Portfolio & Vector Selection (`selectSupplierRnDProject(world, supplier)`)

When a supplier has free engineering capacity and `cash > safetyWageReserve + projectNreEstimate`, its leadership (`CEO` + highest-ranked `Engineer`) scores candidate R&D vectors:

1. **Candidate Directions per Domain:**
    - `arch_leap` (Unlock next interface/node/panel/chemistry tier — high NRE, high failure/rework risk, requires high `analytical`/`openness` & lab equipment).
    - `performance_push` (Push clock/bandwidth/IOPS/megapixels/nits — raises power/cost, appeals to flagship/gamer/photo brands).
    - `efficiency_thermal` (Cut `loadW`, `idleMw`, or `rthCW` — appeals to battery-constrained or thin-phone brands).
    - `yield_cost_shrink` (Reduce `dieAreaMm2` / `D0` / BOM input usage — lowers unit cost so the supplier can undercut rivals on price).

2. **Scoring Formula (100% Grounded in Firm Reality):** $$\text{Score}(v) = w_{\text{skill}}(v) \cdot \text{TeamCapabilityFit}(v) + w_{\text{market}} \cdot \text{ObservedDemandGap}(v) + w_{\text{dna}} \cdot \text{CEO_DNA}(v) - w_{\text{risk}} \cdot \frac{\text{NRE}(v)}{\text{FreeCash}}$$
    - **`TeamCapabilityFit(v)`:** Uses `HumanCore.evaluateAbility()` and `H.fit(profile, spec)` of actual employees currently on payroll. A firm *knows* what its engineers are actually good at!
    - **`ObservedDemandGap(v)`:** Aggregates real telemetry from recent brand BOM searches (`rfqTelemetry`): e.g., if handset brands keep rejecting `chip` quotes because `peakW` exceeds available batteries, `efficiency_thermal` skyrockets in priority!

3.2 R&D Execution with Real Breakthroughs, Stalls & Imperfect Prototypes

- Progress per working hour sums `H.evaluateAbility(p.profile, p.state, skill, spec, { toolMultiplier, complexity })` across assigned engineers.
- At qualification gate, compare achieved prototype quality (`qualityChance` weighted across engineering hours + test equipment) against target tolerances:
    - **Exceptional Roll + High Skill (`> 0.88`): Breakthrough SKU** (`+8%..15%` better efficiency/yield than baseline target), cementing that firm's global reputation in that seed!
    - **Pass (`0.55..0.88`):** Standard qualified SKU enters the firm's commercial catalog.
    - **Marginal Pass (`0.38..0.55`):** Released as a **compromised / B-grade SKU** (e.g., higher wattage or lower clock than hoped—still sellable at a discount!).
    - **Failure (`< 0.38`):** Requires paid rework/respin or cancellation (sunk NRE logged honestly; if cash runway is exhausted, the project stalls or is shelved).

3.3 Demand-Driven Lot Production, Yield Learning & Dynamic Pricing

1. **Smart Lot Scheduling (Replaces Blind `q < 120` Loop):** Suppliers rank their qualified SKUs by `expectedMargin * (recentOrderVelocity + activeBrandCompatibilityCount) - currentDaysOfStock`, allocating raw feed and cleanroom/line hours to SKUs that handset brands actually want—retiring obsolete SKUs once demand drops to zero.
2. **Cumulative Learning Curve (D_0):** Each lot produced on a process increments `processCumulativeUnits`, lowering defect density D_0 toward a physical floor (`wafer()` yield rises from e.g. `52%` on early runs to `86%` on mature runs), rewarding specialist suppliers who stay focused!
3. **Autonomous Bid/Ask Pricing:** Weekly review adjusts SKU quote markup based on **(a)** competitor price-per-performance in the same category, **(b)** own inventory days-on-hand (cutting price down to `marginalCost * 1.04` to clear slow lots; raising price up to scarcity equilibrium when backlogged), and **(c)** CEO `ambition` vs. `prudence`.

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4. Cross-Sector Emergent Ecosystems: How Handset Brands Adapt to Each World

Because handset brands use the **Combinatorial BOM Solver** (see `LIVING_ECOSYSTEM_ENGINE_AND_AUTONOMOUS_DIFF_SPEC`) rather than fixed generation templates, asymmetric supplier worlds automatically create **distinct technological eras and phone metas** in every New Game:

- **Scenario 1 — "Chip Superpower, Weak Battery/Cooling World":** Chip Vendor A invents a monster 4-core high-clock SoC early, while Battery and Enclosure suppliers have weak teams stuck on low-Wh cells and plastic chassis. *Emergent NPC Adaptation:* Ambitious brands that blindly slap the monster chip into plastic phones fail thermal validation (`surfaceC > 50°C`) or suffer severe throttling and RMAs; smart brands either underclock/buy `B-Bin` chips, fund internal thermal/OS optimization, or wait until an enclosure firm finally cracks aluminum-graphite cooling!
- **Scenario 2 — "RAM & Storage Abundance, Stagnant SoC World":** DRAM and NAND firms have star engineers delivering cheap, fast high-capacity memory, while CPU suppliers suffer respin delays. *Emergent NPC Adaptation:* Handset brands compete on **smooth multitasking (`dailyFluidity`), zero-lag storage, massive photo storage tiers (`Base`/`Pro`), and lean OS optimization** rather than raw 3D gaming GFLOPS!
- **Scenario 3 — "Optics Golden Age":** A boutique Camera supplier led by an optics luminary creates a large-sensor OIS module early. *Emergent NPC Adaptation:* Photography consumer cohorts surge in conversion because ΔU over old feature phones is massive, turning camera-focused handset brands into cash-rich market leaders while chip-focused gaming brands struggle in a niche!

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5. Implementation Data Contracts & Multi-Seed Divergence Verification

5.1 Required State Additions (Schema Cleanliness)

```javascript
// Supplier firm extensions in world.firms[]
{
  id: 'supplier-dram-1',
  role: 'dram_memory',           // Canonical specialized role: chip, dram_memory, nand_storage, camera_module, display_panel, battery_pack, enclosure, ems
```

```
  foundingArchetype: 'boutique', // 'boutique' | 'industrial_spinout' |
'commodity_scale' | 'moonshot_lab'
  dna: { performanceBias: 0.22, efficiencyBias: 0.78, costYieldBias: 0.65,
innovationRisk: 0.41 },
  equipmentTier: 1,              // Paid upgradable lab/cleanroom tooling (affects
evaluateAbility toolMultiplier & node access)
  processExperience: { 'lpddr2_65nm': 140 }, // Cumulative lots for D0 yield
learning
  skuCatalog: ['sku_ram_102', 'sku_ram_109'],
  rfqTelemetry: { wins: 14, lostPrice: 3, lostPower: 8, lostBusMismatch: 2 }
}
```

(Note: Split `memory` into distinct `dram_memory` and `nand_storage` supplier roles—at least 2 competing firms per role—so DRAM and Storage evolve independently as requested by the owner!)

5.2 Thai Observer Transparency (Making World History Readable)

Add an **"ภาพรวมมหาอำนาจชิ้นส่วนของโลกนี้ (World Industry Balance)"** summary banner in the Supply/Market tab computed purely from live SKU benchmarks:

- Show which component sector currently leads technological advancement in this seed, who the star engineers (`Luminaries`) are, which firm holds the highest market share per part category, and which sector is currently the **industry-wide bottleneck** (e.g., *"โลกนี้วงการหน่วยความจำ DRAM ก้าวหน้าเร็วจากทีมวิจัยของ ธาราเมมโมรี แต่อุตสาหกรรมแบตเตอรี่ยังติดคอขวดความจุ ทำให้มือถือส่วนใหญ่ต้องเน้นประหยัดพลังงาน"*).

5.3 Mandatory Automated Tests (`SPINOFF_SOURCE/test/simulation.test.cjs`)

1. **Cross-Seed Sector Divergence Test:** Run 6 distinct seeds (`seed-1 .. seed-6`) for `365` days. Assert that the **rank-ordering of sector technological progress** (`chip` vs. `dram` vs. `storage` vs. `camera` vs. `battery` vs. `display`) is **NOT identical across all 6 seeds**, proving genuine world-to-world replayability!
2. **Causal NPC-Driven Advantage Test:** Take two identical worlds except swap a high-aptitude specialist team from `supplier-ram-1` to `supplier-camera-1` at `minute = 0` (keeping cash conserved). Run `240` days and assert that `camera` SKUs advance further in World B while `ram` SKUs advance further in World A.
3. **Zero Scripted Buff / Conservation Audit:** Verify `audit(world).ok === true` daily across all seeds: all supplier equipment upgrades, NRE, raw feeds, and wages reconcile to the exact satang, `human-core.js` SHA-256 matches `CORE_LOCK.json`, and `advance(w, 60)` equals `12 × advance(w, 5)`.